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Idiosyncratic Jump Risk Matters: Evidence from Equity Returns and Options

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1 Big picture

- **Question.** Does idiosyncratic risk carry an equity premium, and is the premium driven by Gaussian volatility or by jump/tail risk?
- **Contribution.** The paper uses option prices to decompose expected stock excess returns into systematic normal, systematic jump, idiosyncratic normal, and idiosyncratic jump premia.
- **Main result.** Idiosyncratic risk explains about 28% of the variation in an average stock's expected excess return, and the contribution comes essentially from idiosyncratic jump risk.
- **Economic takeaway.** Total idiosyncratic volatility is a noisy proxy because the priced part is the tail component, not ordinary Gaussian idiosyncratic volatility.

2 Data and measurement

The sample uses daily equity returns and weekly cross-sections of out-of-the-money options on the S&P 500 index and 260 firms from January 1996 to August 2015. The key objects are systematic normal risk, systematic jump risk, idiosyncratic normal risk, and idiosyncratic jump risk. Option data are central because they reveal information about skewness, kurtosis, jump intensity, and risk-neutral compensation that is difficult to identify from returns alone.

Interpretation: The paper's empirical object is an option-implied risk-premium decomposition. The most important measurement distinction is between idiosyncratic Gaussian risk, which is diversifiable, and idiosyncratic jump risk, which is hard to hedge and partly common in bad states.

3 Models and empirical specifications

3.1 Return decomposition

$$\begin{aligned} r_{M,t+1} &= \mu_{M,t+1} - \xi_{M,t+1}^P + z_{M,t+1} + y_{M,t+1}, \\ r_{S,t+1} &= \mu_{S,t+1} - \xi_{S,t+1}^P + \beta_{S,z} z_{M,t+1} + \beta_{S,y} y_{M,t+1} + z_{S,t+1} + y_{S,t+1}. \end{aligned}$$

Interpretation: Stocks are exposed to market Gaussian and jump shocks, plus stock-specific Gaussian and jump shocks.

3.2 Equity premium decomposition

$$ERP_{S,t} = SNRP_{S,t} + SJRP_{S,t} + INRP_{S,t} + IJRP_{S,t}.$$

Interpretation: The decomposition identifies whether the expected return is compensation for systematic risk, idiosyncratic normal risk, or idiosyncratic jump risk.

3.3 Portfolio tests

$$R_t^{LS} = \alpha + \beta' F_t + \varepsilon_t.$$

Long-short portfolios are formed by sorting on INRP or IJRP. A positive alpha for IJRP portfolios means that option-implied idiosyncratic jump exposure predicts returns beyond standard factors.

4 Identification and empirical design

The paper is structural rather than a natural experiment. Identification comes from joint restrictions imposed by returns, options, no-arbitrage pricing, the market-first estimation structure, and the clearing condition that idiosyncratic premia net out across the market portfolio.

- Out-of-the-money options identify tail probabilities and jump-risk compensation.
- Cross-sectional variation in moneyness and maturity separates volatility, persistence, jump intensity, and jump-size parameters.
- Market index returns and options identify systematic normal and jump factors.
- Stock returns and single-stock options identify firm-specific normal and jump innovations conditional on market shocks.
- The nested model without idiosyncratic jumps is a falsification test: it produces excessive residual kurtosis and implausible ERP dispersion.
- Factor regressions, double sorts, sample-representativeness checks, and short-term-option estimates test whether the IJRP result is driven by missing standard factors, firm characteristics, sample selection, or option maturity.

Interpretation: The most credible evidence is the combination of model fit, nested-model failure, and return-predictability tests. The limitation is that the decomposition is model-based.

5 Tables and figures

5.1 Table 1: Market parameters estimated using returns and option data; Figure 1: Annualized normal and jump risk premiums for the market model

Read: Table 1 estimates market normal and jump parameters. Figure 1 shows the time series of the market ERP and the share explained by jumps.

Economic message: Market jump risk is a large part of aggregate compensation, especially during turmoil.

Detailed interpretation: The market model provides the systematic benchmark used for all stock-level decompositions.

Table 1
Market parameters estimated using returns and option data

	Normal	Jump
λ_M / γ_M	8.24×10^{-1} (2.40×10^{-10})	7.01×10^{-1} (3.55×10^{-10})
$w_{M,z} / w_{M,y}$	-1.63×10^{-6} (2.50×10^{-8})	-4.05×10^{-7} (1.88×10^{-8})
$a_{M,z} / a_{M,y} \times 10^6$	2.42×10^0 (3.36×10^{-3})	4.44×10^0 (7.83×10^{-3})
$b_{M,z} / b_{M,y}$	9.40×10^{-1} (6.94×10^{-10})	9.34×10^{-1} (4.58×10^{-9})
$c_{M,z} / c_{M,y}$	1.44×10^2 (2.86×10^{-11})	1.41×10^2 (1.30×10^{-11})
α_M		1.19×10^1 (1.31×10^{-08})
δ_M		-7.02×10^0 (3.01×10^{-10})
Skewness of innovations, $\varepsilon_{M,t} / \gamma_{M,t}$	-0.12	-7.37
Total skewness, $z_{M,t} + \gamma_{M,t}$	-0.31	346.33
Ex. kurtosis of innovations, $\varepsilon_{M,t} / \gamma_{M,t}$	-0.07	7.75
Total ex. kurtosis, $z_{M,t} + \gamma_{M,t}$	74.0	26.0
% of annual variance	15.6	9.0
Average volatility		
Total		18.0
Average risk premium (Equation (10))	2.33	3.85
Total	6.18	
Median risk premium (Equation (10))	1.72	3.04
Total	4.75	
Return log-likelihood	78,395.3	
Option log-likelihood	12,787.7	
Total log-likelihood	91,183.0	
RIVRMSE (Equation (25))	14.4	

(a) Table 1: Market parameters estimated using returns and option data.

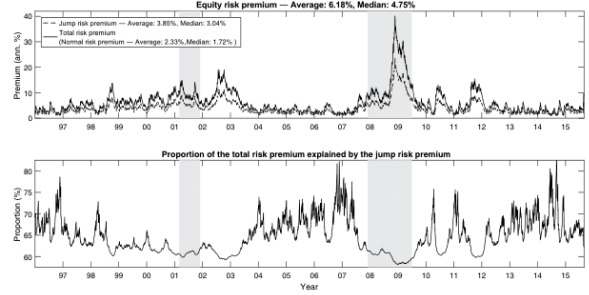


figure 1
annualized normal and jump risk premiums for the market model
The top panel of this figure reports the annualized equity risk premium, $252(\lambda_M h_{M,z,t} + \gamma_M h_{M,y,t})$ and the component due to jump risk, $252\gamma_M h_{M,y,t}$. The lower panel reports, on a daily basis, the proportion of the total risk premium explained by the jump risk premium.

5.2 Table 2: Valuation errors; Table 3: Firm parameters estimated using returns and option data

Read: Table 2 reports option fit. Table 3 reports the central estimates: average absolute IJRP is 2.44%, while average absolute INRP is nearly zero.

Economic message: The priced part of idiosyncratic risk is the jump component.

Detailed interpretation: Table 3 also shows strong commonality in idiosyncratic jump risk, explaining why tail risk is difficult to diversify in bad states.

5.3 Figure 2: Time-series decomposition; Figure 3: Cross-sectional decomposition

Read: These figures show the four premium components over time and across firms.

Economic message: The idiosyncratic jump component is widespread, time-varying, and economically meaningful.

5.4 Figure 4: Commonality; Figure 5: Decomposition neglecting idiosyncratic jumps

Read: Figure 4 documents factor structure in shocks to idiosyncratic risk. Figure 5 shows what happens when idiosyncratic jumps are removed.

Economic message: The nested model misclassifies tail events as Gaussian idiosyncratic risk.

Table 2
Valuation errors on the options used for the estimation

A. Market

Overall IVRMSE and RIVRMSE			Sorting by year		
	IVRMSE	RIVRMSE		IVRMSE	RIVRMSE
All	3.086	14.391	1996	2.326	15.834
Sorting by maturity			1997	3.844	17.178
			1998	5.318	20.129
			1999	4.625	17.786
			2000	2.419	10.508
			2001	2.756	11.757
			2002	2.192	9.345
			2003	2.313	11.214
DTM ≤ 30	3.364	16.185	2004	1.260	8.067
30 < DTM ≤ 90	3.108	14.606	2005	2.307	17.593
90 < DTM ≤ 180	2.936	13.590	2006	2.327	17.721
180 < DTM ≤ 270	3.016	13.607	2007	2.002	14.031
270 < DTM ≤ 365	3.076	14.137	2008	5.086	13.635
Sorting by moneyness			2009	4.018	15.546
			2010	3.724	17.210
			2011	3.416	13.978
			2012	2.615	13.138
			2013	1.255	8.961
			2014	1.543	11.914
			2015	2.018	16.756
0.80 < $K/F \leq 0.85$	3.815	11.663			
0.85 < $K/F \leq 0.90$	3.533	11.818			
0.90 < $K/F \leq 0.95$	3.334	12.543			
0.95 < $K/F \leq 1.00$	3.006	13.421			
1.00 < $K/F \leq 1.05$	2.684	16.316			
1.05 < $K/F \leq 1.10$	2.729	16.577			
1.10 < $K/F \leq 1.15$	2.808	15.384			
1.15 < $K/F \leq 1.20$	3.037	15.134			

B. Stocks

Overall average IVRMSE and RIVRMSE			Sorting by year		
	IVRMSE	RIVRMSE		IVRMSE	RIVRMSE
All	7.355	17.239	1996	4.525	13.321
Sorting by maturity			1997	5.158	13.635
			1998	6.480	13.861
			1999	6.642	13.673
			2000	9.382	16.411
			2001	7.815	15.557
			2002	7.878	16.872
			2003	5.429	14.791
DTM ≤ 30	9.799	19.115	2004	4.091	13.741
30 < DTM ≤ 90	7.223	16.459	2005	4.187	15.141
90 < DTM ≤ 180	6.590	16.464	2006	4.450	14.870
180 < DTM ≤ 270	6.762	17.762	2007	5.320	16.235
270 < DTM ≤ 365	7.032	19.163	2008	11.721	18.163
Sorting by moneyness			2009	9.237	16.932
			2010	4.903	14.374
			2011	5.669	15.280
			2012	4.811	14.736
			2013	3.913	14.119
			2014	4.340	16.374
			2015	4.479	16.759
0.80 < $K/F \leq 0.85$	9.957	20.494			
0.85 < $K/F \leq 0.90$	8.494	17.856			
0.90 < $K/F \leq 0.95$	7.179	15.908			
0.95 < $K/F \leq 1.00$	6.523	16.177			
1.00 < $K/F \leq 1.05$	6.127	16.099			
1.05 < $K/F \leq 1.10$	6.667	15.853			
1.10 < $K/F \leq 1.15$	7.805	18.945			
1.15 < $K/F \leq 1.20$	9.104	21.412			

We use the joint MLE estimates from Tables 1 and 3 to compute the implied volatility root-mean-square errors

Table 3 Firm parameters estimated using returns and option data	Average	SD	Min	Q1	Median	Q3	Max
Based on 260 firms:							
$\beta_{E,t}$	0.931	0.277	0.320	0.730	0.935	1.108	1.75
$\beta_{E,t}$	1.111	0.412	0.206	0.831	1.080	1.315	3.921
$\lambda_{E,t}$	0.000	0.002	-0.017	-0.000	0.000	0.000	0.01
$\gamma_{E,t}$	2.021	0.588	0.496	1.632	2.022	2.363	4.241
$\kappa_{E,t}$	0.803	0.086	0.057	0.402	0.870	1.110	3.93
$\kappa_{E,t} \times 10^6$	1.920	1.344	0.113	1.007	1.574	2.473	9.65
$\beta_{E,t}$	0.992	0.003	0.978	0.990	0.992	0.994	1.000
$\beta_{E,t}$	97.56	47.14	-87.97	60.81	104.77	133.80	209.16
$\kappa_{E,t}$	0.335	0.321	0.091	0.103	0.217	0.460	2.20
$\kappa_{E,t} \times 10^6$	4.301	3.044	0.454	1.995	3.929	5.939	28.22
$\beta_{E,t}$	0.950	0.059	0.325	0.909	0.935	0.965	0.996
$\kappa_{E,t}$	19.49	52.56	-262.49	101.52	126.53	141.30	381.37
$\beta_{E,t}$	10.655	4.204	0.764	8.779	10.705	12.398	43.83
$\lambda_{E,t}$	-8.263	2.200	-15.334	-7.528	-6.517	-5.243	-0.33
Proportion of variation in F explained by X ($F \sim X$):							
$\text{Var}_t^2(\varepsilon_{E,t+1}) \sim \text{Common idio. normal risk (\%)}$	42.27	35.07	0.47	9.28	29.66	79.60	100.00
$\text{Var}_t^2(\varepsilon_{E,t+1}) \sim \text{Common idio. jump risk (\%)}$	56.38	36.66	0.00	17.51	63.56	93.40	99.98
$\text{Var}_t^2(\varepsilon_{E,t+1} + \varepsilon_{E,t+1}^2) \sim \text{Common idiosyncratic risk (\%)}$	42.08	35.14	0.01	9.50	30.49	78.48	100.00
$\text{ERP} \sim \text{ERP (\%)}$	28.17	19.93	0.02	12.01	22.79	40.06	84.42
Skewness of Gaussian innovations, $\varepsilon_{E,t}$:							
Skewness of innovations, $\varepsilon_{E,t} + \varepsilon_{E,t}^2$	0.11	0.28	-2.48	0.07	0.13	0.19	1.44
Skewness of innovations, $\varepsilon_{E,t} + \varepsilon_{E,t}^2$	-0.97	3.91	-33.46	-0.71	-0.11	0.13	2.48
Ex. kurtosis of Gaussian innovations, $\varepsilon_{E,t}$	1.52	4.21	-0.08	0.65	0.93	1.20	8.99
Ex. kurtosis of innovations, $\varepsilon_{E,t} + \varepsilon_{E,t}^2$	46.02	172.07	1.93	6.77	11.38	26.47	153.00
RIVRMSE	17.24	5.61	9.50	13.43	16.38	19.14	59.23
Distribution of 260 weighted cross-sectional average:							
Total volatility (%)	11.80	8.48	16.26	24.94	29.58	37.08	62.54
Idio. variance / Total variance (%)	49.23	8.62	47.45	62.94	70.26	75.58	90.39
Idio. jump variance / Idio. variance (%)	9.50	1.18	6.05	8.62	9.49	10.35	13.66
Absolute value of ERP (%)	0.01	0.09	0.00	0.01	0.01	0.01	0.02
Absolute value of ERP (%)	2.44	1.09	0.92	1.71	2.12	2.82	8.96

Parameters for each stock are estimated using daily index returns and available weekly cross-sections of out-of-the-money options, from January 1996 to August 2015. This table reports statistics on the cross-section (260 stocks) of joint MLE estimates obtained for the 260 stocks in our sample. The table further documents the distributions of model-implied measures of volatility, risk premiums, and fit to the data. Q1 and Q3 report the 25th and 75th percentiles of the estimates. The proportion of conditional variance, $\text{Var}_t^2(\varepsilon_{E,t+1} + \varepsilon_{E,t+1}^2)$, explained by common idiosyncratic risk is obtained using the formula in Equation (77). Other measures are obtained using variations of this formula. Finally, the lower part of the table reports

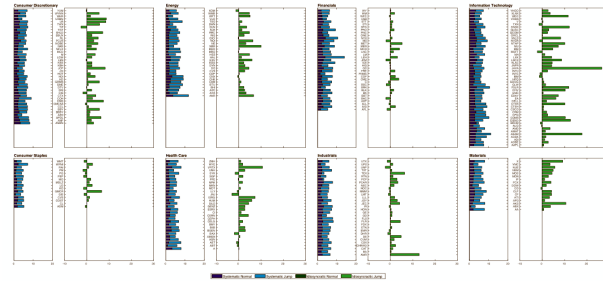
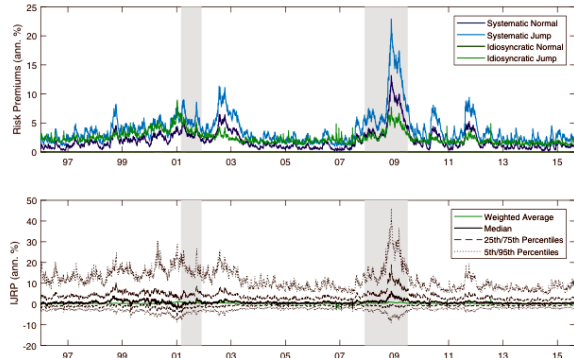
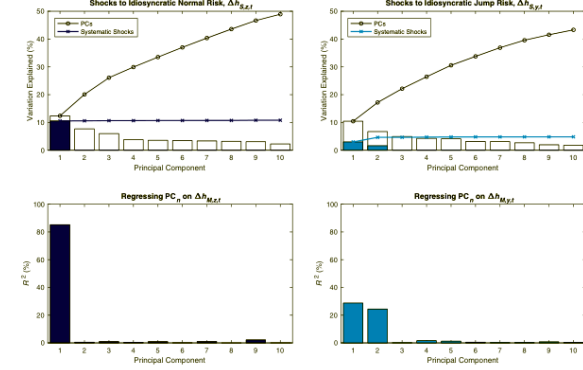


Figure 3
Decomposition of the equity risk premium on stocks: Cross-section of time-series averages
This figure presents, for each of the 260 stocks, the decomposition of the equity risk premium in terms of the premiums associated with the four different risk factors in the model



(a) Figure 4: Commonality of idiosyncratic risk shocks.

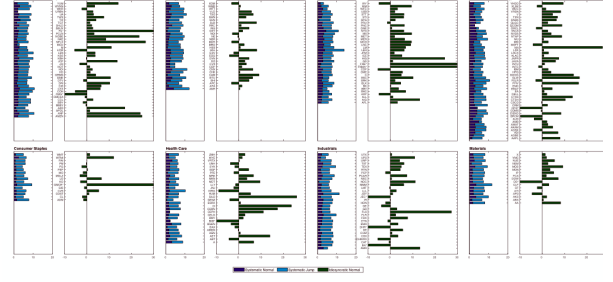


Figure 5
Decomposition of the equity risk premium by firm: Neglecting idiosyncratic jumps
This figure presents, for each of the 260 stocks, the decomposition of the equity risk premium in terms of the premiums associated with the three different risk factors in the nested model Equation (28): (1) systematic normal, (2) systematic jump, and (3) idiosyncratic normal. On average, the premium on (1) systematic normal risk is 1.8%, (2) systematic jump risk, 4.5%, and (3) idiosyncratic normal risk, 1.7%.

(b) Figure 5: Decomposition neglecting idiosyncratic jumps.

5.5 Figure 6: With and without idiosyncratic jumps; Table 4: Parameters neglecting idiosyncratic jumps

Read: The full model gives plausible positive ERPs; the nested model produces excessive dispersion and high residual kurtosis.

Economic message: Omitting idiosyncratic jumps creates model misspecification.

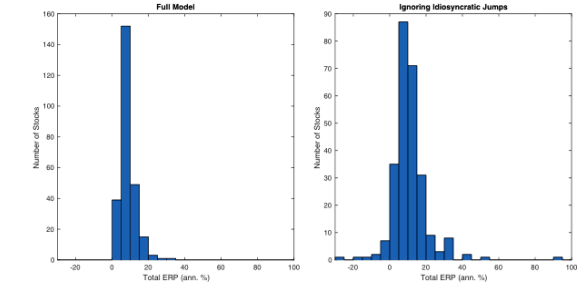


Figure 6
Comparison of the equity risk premium by firm: With and without idiosyncratic jumps
This figure presents the distribution of the equity risk premium on the 260 firms in our sample for the full model (on the left) and the nested model without idiosyncratic jumps (on the right).

(a) Figure 6: ERP with and without idiosyncratic jumps.

	Average	S.D.	Min	Q1	Median	Q3	Max
Based on 260 firms:							
$\beta_{S,z}$	0.879	0.318	0.163	0.649	0.872	1.087	2.060
$\beta_{S,y}$	1.456	0.516	0.001	1.164	1.479	1.747	3.958
λ_S	1.789	1.122	-2.318	0.907	1.708	2.488	4.997
$\kappa_{S,z}$	0.829	0.779	0.045	0.339	0.592	1.122	5.797
$\sigma_{S,z} \times 10^6$	2.370	2.173	0.005	1.025	1.811	3.203	22.687
$\beta_{S,z}$	0.996	0.011	0.910	0.998	0.999	0.999	1.000
$\epsilon_{S,z}$	46.71	92.58	-463.50	-2.84	46.05	81.78	530.90
Skewness of Gaussian innovations, $\epsilon_{S,t}$	-0.37	1.13	-8.14	-0.56	-0.08	0.11	2.08
Skewness of innovations, $z_{S,t} + y_{S,t}$	-3.29	2.33	-15.97	-4.57	-3.15	-1.58	0.00
Ex. kurtosis of Gaussian innovations, $\epsilon_{S,t}$	16.46	28.55	0.65	4.99	9.35	16.23	292.95
Ex. kurtosis of innovations, $z_{S,t} + y_{S,t}$	169.97	156.17	0.00	57.60	145.02	234.67	1265.67
RIVRMSE	15.33	6.41	9.94	12.20	13.73	16.50	63.90
Distribution of 260 weighted cross-sectional average:							
Total volatility (%)	36.44	9.53	18.36	29.16	34.85	43.32	63.69

This table reports statistics on the joint MLE estimates obtained using a model neglecting firm-specific jumps.

(b) Table 4: Parameters when idiosyncratic jumps are neglected.

5.6 Table 5: Exposure to factors; Figure 7: Representativeness of the sample

Read: These visuals compare the 260-stock option sample with the CRSP universe and with standard factor exposures.

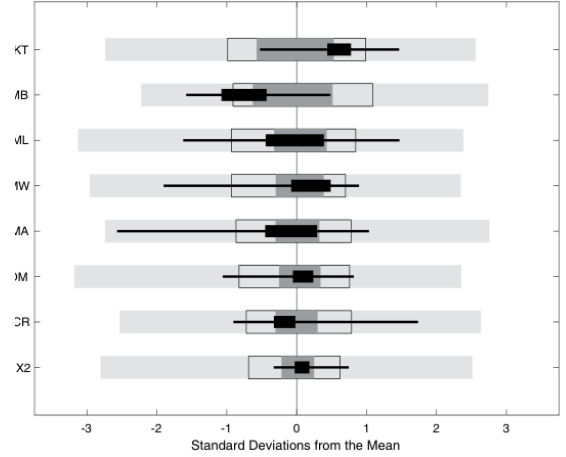
Economic message: The sample is large-liquid-firm biased but reasonably representative in risk exposures.

Table 5
Exposure to factors

	CRSP	Sample	Model decomposition			
			SNR	SJR	INR	IJR
MKT	0.970	1.000	0.748	0.100	0.095	0.004
	[0.50]	[0.89]	[1.00]	[0.72]	[0.27]	[0.04]
SMB	0.010	-0.189	-0.115	-0.006	-0.045	-0.004
	[0.32]	[0.23]	[0.87]	[0.29]	[0.26]	[0.04]
HML	0.010	-0.089	-0.003	0.006	-0.075	-0.001
	[0.17]	[0.33]	[0.11]	[0.04]	[0.34]	[0.06]
RMW	0.004	0.107	0.088	-0.027	0.032	0.001
	[0.13]	[0.27]	[0.66]	[0.15]	[0.28]	[0.05]
CMA	-0.013	-0.037	0.006	-0.028	-0.052	-0.019
	[0.12]	[0.30]	[0.25]	[0.03]	[0.30]	[0.05]
MOM	-0.009	-0.003	-0.030	0.020	0.005	-0.001
	[0.17]	[0.32]	[0.39]	[0.12]	[0.34]	[0.07]
ICR	0.013	0.002	0.006	-0.009	0.005	-0.000
	[0.12]	[0.15]	[0.04]	[0.08]	[0.15]	[0.03]
ΔVIX^2	-0.015	-0.010	0.023	-0.036	0.006	-0.001
	[0.10]	[0.09]	[0.17]	[0.19]	[0.10]	[0.03]
R^2	.181	.364	.904	.456	.159	.045

To obtain the first column of this table, on each day t , for each stock in the CRSP universe, we perform a multivariate regression of the stock's returns on the set of factors using observations on the last 252 trading days. For each loading, a cross-sectional average is computed using market capitalization weights, and we report the time-series average of these cross-sectional averages. In square brackets, we report the corresponding proportion of loadings that are statistically significant at the 5% level in those regressions, based on Newey and West (1987) standard errors. We finally report the average R^2 . The Sample column report the same results, restricting the

(a) Table 5: Exposure to factors.



(b) Figure 7: Representativeness of the sample.

5.7 Table 6: Zero-investment long-short portfolios based on idiosyncratic risk premiums

Read: IJRP-sorted long-short portfolios earn large positive alphas across factor models; INRP portfolios are much weaker.

Economic message: The return tests support the option-implied conclusion that idiosyncratic jump risk is priced.

Cat	Zero-investment long-short portfolios based on idiosyncratic risk premiums									
	Idiosyncratic diffusive risk premium					Idiosyncratic jump risk premium				
	FF3	FF5	MOM	HKM	AHGX	FF3	FF5	MOM	HKM	AHGX
MKT	-0.93 (-0.35)	1.66 (0.62)	0.49 (0.18)	-2.04 (-0.75)	-0.53 (-0.20)	2.44 (0.93)	9.24** (2.82)	13.64** (4.26)	11.95** (3.77)	9.68** (2.98)
SMB	0.36** (15.28)	0.27** (9.50)	0.32** (14.52)	0.46** (18.46)	0.13** (12.64)	0.38** (10.68)	1.17** (43.39)	1.02** (43.18)	1.11** (50.02)	1.12** (41.28)
HML	0.19** (5.80)	0.15** (4.43)	0.21** (6.05)	0.20** (6.29)	0.15** (5.28)	0.08** (2.33)	0.02 (0.09)	0.11** (3.60)	0.08** (2.60)	0.05** (1.75)
RMW	-0.26** (-7.71)	-0.11** (-3.20)	-0.34** (-10.08)	-0.28** (-8.39)	-0.12** (-3.26)	-0.09 (-1.49)	0.21** (5.11)	-0.25** (-6.68)	-0.11** (-3.70)	0.07** (2.07)
CMA	-0.21** (-6.54)	-0.21** (-6.54)	-0.21** (-6.54)	-0.21** (-6.54)	-0.21** (-6.54)	-0.21** (-6.54)	-0.21** (-6.54)	-0.21** (-6.54)	-0.21** (-6.54)	-0.21** (-6.54)
MOM			-0.16** (-5.19)		-0.13** (-5.79)			-0.30** (-7.79)		-0.22** (-6.40)
ICR			-0.10** (-6.33)		-0.10** (-6.33)			0.01 (0.43)		-0.08** (-3.28)
ΔVIX^2				-0.07** (-3.09)	-0.09** (-4.10)				-0.07** (-3.13)	-0.10** (-4.12)
Adj. R^2	.27	.31	.29	.24	.28	.34	.69	.74	.69	.75

Table 6: Zero-investment long-short portfolios.

5.8 Table 7: Quintile portfolio characteristics; Table 8: Double-sort alphas

Read: Table 7 shows characteristics of IJRP quintiles; Table 8 shows that alphas survive double sorts.

Economic message: IJRP is not simply beta, size, value, profitability, investment, momentum, ICR beta, or variance beta.

Table 7
Description of quintile portfolios

	Idiosyncratic jump risk premium				
	P1	P2	P3	P4	P5
Market beta	0.95 [0.07]	0.99 [0.07]	1.04 [0.09]	1.06 [0.10]	1.08 [0.14]
log(ME)	25.30 [0.32]	25.03 [0.35]	24.72 [0.46]	24.43 [0.48]	24.07 [0.59]
BE/ME	1.01 [0.34]	1.12 [0.34]	1.24 [0.68]	1.24 [0.82]	1.50 [1.48]
OP (%)	5.79 [1.42]	5.49 [1.57]	5.51 [1.40]	5.46 [1.53]	5.12 [1.90]
Investment (%)	16.34 [22.35]	15.95 [17.71]	19.31 [26.10]	22.01 [31.13]	29.82 [58.86]
Return [-12,-2] (%)	14.89 [18.03]	15.26 [18.18]	17.33 [22.77]	19.09 [26.77]	23.41 [38.41]
ICR beta	3.32 [5.02]	0.24 [4.17]	-0.84 [4.68]	-2.56 [4.93]	-4.69 [6.75]
Variance beta	-3.80 [8.05]	-0.85 [6.93]	0.83 [8.34]	1.93 [10.58]	2.92 [14.13]

This table describes the quintile portfolios obtained in Table 6. Each day, the following variables are recorded for the firms in each quintile portfolio: market beta, log of market capitalization, book-to-market ratio, operating profitability (OP), investment, trailing 12-month return, and the volatility beta. This table reports the time-series

(a) Table 7: Description of quintile portfolios.

Table 8
Double sort: Alphas of portfolios based on idiosyncratic jump risk premiums

	Q1	Q2	Q3	Q4	Q5
Market beta	10.26** (2.61)	19.44** (5.16)	13.43** (3.56)	12.85** (3.09)	15.47** (3.21)
log(ME)	13.19** (2.74)	9.46** (2.53)	10.87** (3.40)	18.62** (5.56)	7.59** (3.06)
BE/ME	16.01** (3.93)	12.14** (2.79)	13.02** (3.17)	12.28** (2.75)	9.19** (2.39)
OP (%)	13.58 (1.50)	12.29** (2.77)	11.17** (2.60)	7.86* (1.83)	16.12** (3.62)
Investment (%)	11.40** (2.73)	13.57** (3.89)	12.79** (3.43)	14.35** (3.68)	13.31** (2.62)
Return [-12,-2] (%)	9.30* (1.85)	12.08** (3.54)	13.52** (3.85)	16.44** (4.85)	12.69** (2.40)
ICR beta	16.85** (3.69)	15.84** (4.05)	11.18** (3.04)	16.25** (3.89)	9.44** (2.15)
Variance beta	9.73** (2.37)	11.85** (3.25)	13.06** (3.55)	15.81** (3.67)	19.40** (3.81)

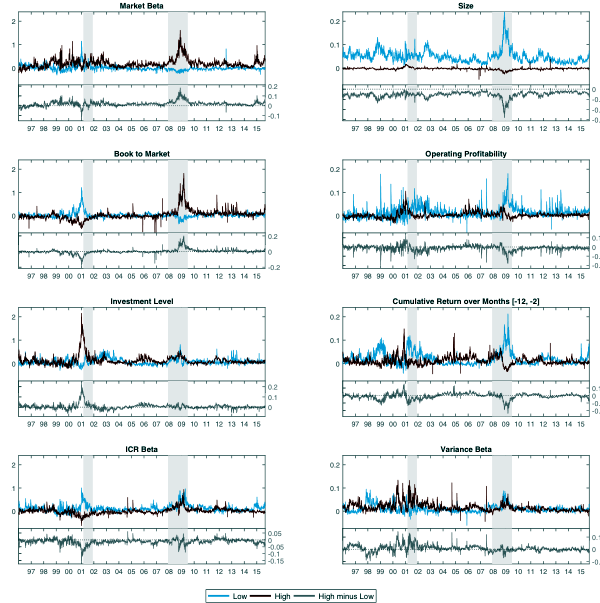
On each day t , we first sort stocks into quintiles (Q1 to Q5) based on their market beta over the past year (cf. Table 5). Then, within each market-beta quintile, we sort stocks into terciles based on $RP_{5,y,t}$. We then take a long position in a cap-weighted portfolio of the top-tercile stocks and a short position in a cap-weighted portfolio of the bottom-tercile stocks. This leaves us with five long-short portfolios, each of which is composed of stocks with homogeneous market betas. The first row of this table reports the alphas of regression (29) for each of these long-short portfolios. The procedure is repeated for seven other variables: log of market capitalization, book-to-

(b) Table 8: Double-sort alphas.

5.9 Figure 8: Stock characteristics and IJRP; Figure 9: IJRP against market equity

Read: These figures show that size is related to IJRP but does not explain all IJRP variation.

Economic message: The premium is related to size but not reducible to size.



(a) Figure 8: Stock characteristics and IJRP.

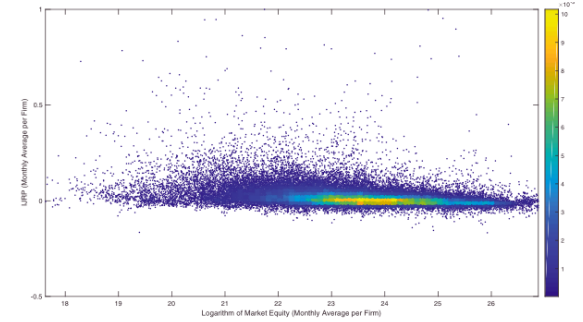


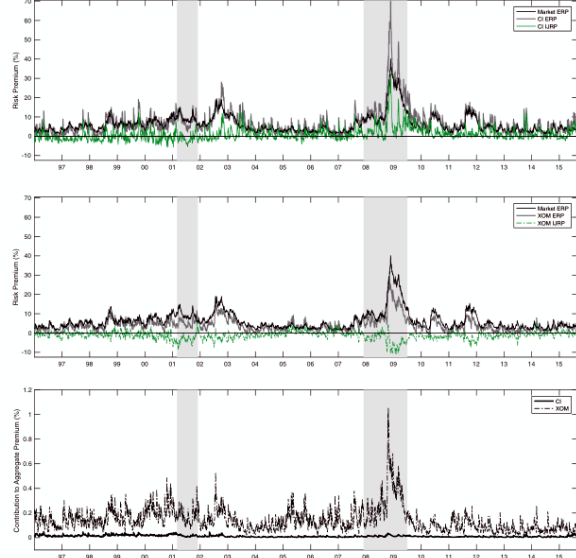
Figure 9
idiosyncratic jump risk premium against market equity
For each firm, the monthly average of the logarithm of the market equity and the idiosyncratic jump risk premium are calculated. The figure then presents a scatterplot of the relationship of the averaged idiosyncratic jump risk premium against the monthly average of the logarithm of the market equity. The color scheme represents the density of points in each bin, yellow being the most populated bins and blue the least populated bins.

(b) Figure 9: IJRP against market equity.

5.10 Figure 10: Cigna versus Exxon Mobil; Figure 11: Short-term options versus full sample

Read: Figure 10 illustrates the size/IJRP relation using two firms. Figure 11 shows short-maturity estimates are close to full-sample estimates.

Economic message: The main result is robust to focusing on the options most informative about jumps.



(a) Figure 10: Comparing the largest firm with a relatively small firm.

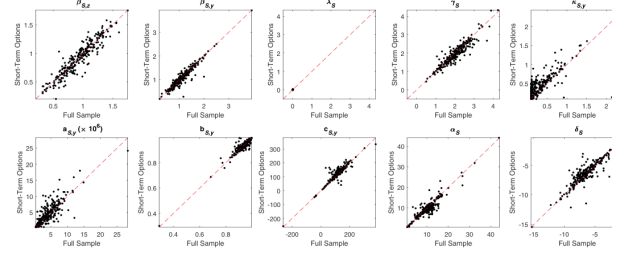


Figure 11

(b) Figure 11: Short-term options versus full sample.

6 Short summary

- Idiosyncratic risk explains about 28% of average stock ERP variation.
- The contribution comes from idiosyncratic jump risk, not idiosyncratic normal risk.
- Commonality in idiosyncratic jump risk is strong.
- Ignoring idiosyncratic jumps creates misspecification and implausible ERPs.
- IJRP long-short portfolios earn large positive alphas after standard controls.

7 Conclusion

7.1 Core conclusion

Idiosyncratic risk matters for expected equity returns only after separating tail risk from Gaussian risk.

7.2 Economic intuition

Gaussian firm-specific risk diversifies, but firm-specific tail risk becomes hard to diversify in market stress.

7.3 Methodological contribution

Option prices allow the authors to identify tail-risk premia that returns alone would miss.

7.4 Takeaway

Idiosyncratic volatility is too coarse. The priced object is idiosyncratic jump risk.